



## Features of the H3 Ball Bearing Rotary Shaft Encoder



- Shafted ball bearing rotary encoder
- 20 Resolutions from 64 to 10,000 CPR (256 to 40,000 PPR)
- 1/4 in. diameter brass shaft mounted in 2 ball bearings
- 2 channel quadrature output with an optional index channel
- High retention connector/cable (sold separately)

## Description

The H3 ball-bearing rotary encoder converts shaft position or speed into a quadrature output signal. This rotary encoder delivers up to 40,000 PPR resolution in a glass-filled polymer housing. The brass shaft rotates on a pair of ball bearings.



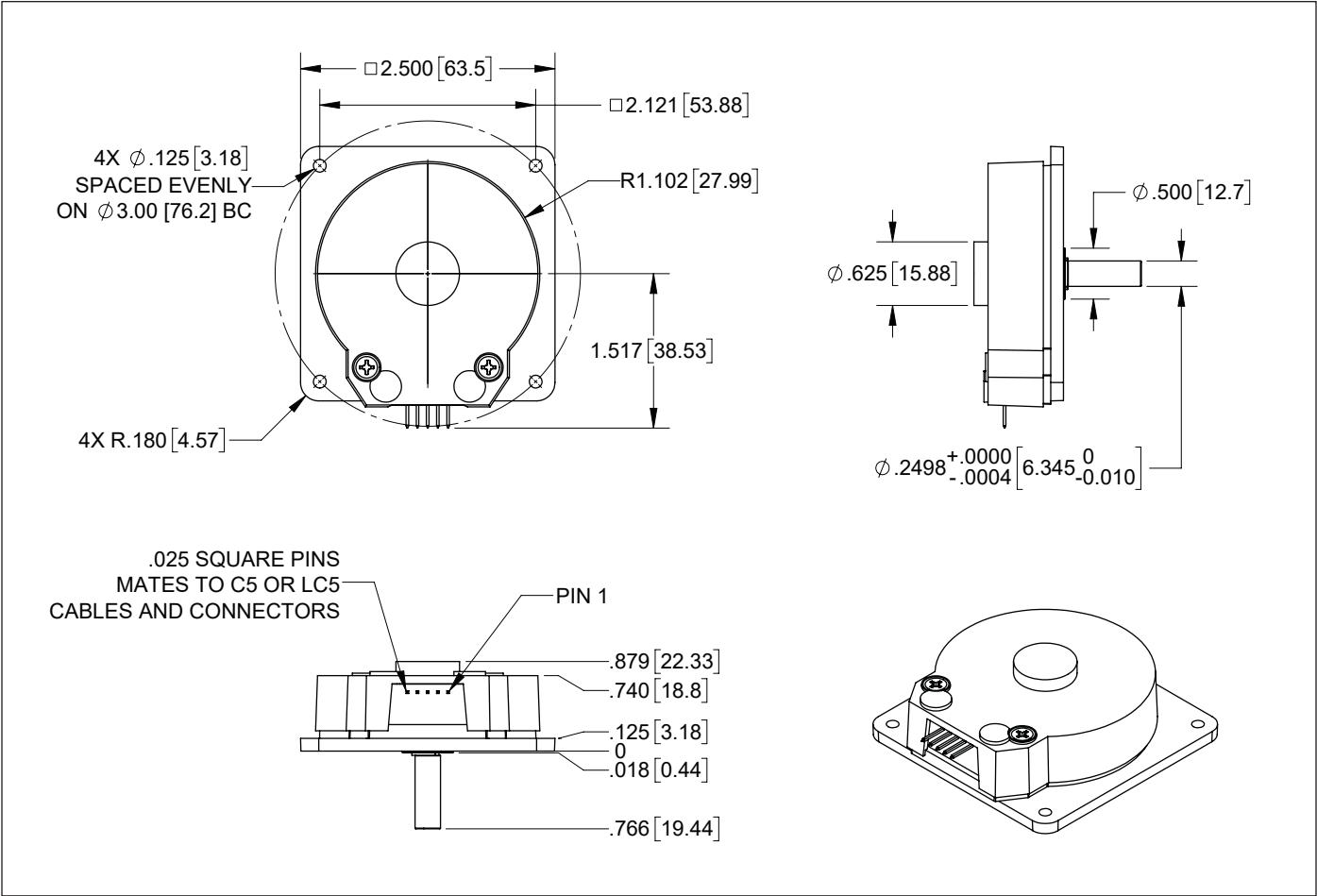
The H3 optical rotary encoder is designed to drive cables up to 10 feet long. For longer cable lengths, we recommend using the model H6<sup>(1)</sup> and selecting the differential output. Another option is to add a PC4<sup>(2)</sup> / PC5<sup>(3)</sup> differential line driver to this encoder.

# Ball Bearing Rotary Shaft Encoder

Product ID | H3



## Mechanical Drawings



H3 Ball Bearing Rotary Shaft Encoder

# Ball Bearing Rotary Shaft Encoder

Product ID | H3



## Specifications

### Environmental

| Parameter                              | Value      | Units |
|----------------------------------------|------------|-------|
| Operating Temperature, CPR < 2000      | -40 to 100 | C     |
| Operating Temperature, CPR ≥ 2000      | -25 to 100 | C     |
| Electrostatic Discharge, IEC 61000-4-2 | ± 4        | kV    |
| Vibration (10Hz to 2kHz, sinusoidal)   | 20         | G     |
| Shock (6 milliseconds, half-sine)      | 75         | G     |

### Mechanical

| Parameter                                             | Value                                                                     |
|-------------------------------------------------------|---------------------------------------------------------------------------|
| Max. Acceleration                                     | 100000 rad/sec <sup>2</sup>                                               |
| Max. Shaft Speed (mechanical)                         | 10000 RPM (1)                                                             |
| Max. Shaft Torque                                     | 0.05 in-oz                                                                |
| Max. Shaft Loading                                    | 2 lbs.                                                                    |
| Bearing Life                                          | life in millions of revs = $(90/P)^3$<br>where P = radial load in pounds. |
| Weight                                                | 2.69 oz.                                                                  |
| Max. Shaft Runout                                     | 0.006 in. T.I.R.                                                          |
| Mounting Plate Screw Torque                           | (#2-56) 2-3                                                               |
| Moment of Inertia                                     | 0.001 oz-in-s <sup>2</sup>                                                |
| Technical Bulletin TB1001 - Shaft and Bore Tolerances |                                                                           |
| <a href="#">Download<sup>(4)</sup></a>                |                                                                           |

(1) The maximum speed due to electrical considerations is dependent on the CPR. See the [EM1<sup>\(5\)</sup>](#) and [EM2<sup>\(6\)</sup>](#) product pages.

# Ball Bearing Rotary Shaft Encoder

Product ID | H3



## Specifications

### Phase Relationship

B leads A for clockwise shaft rotation, and A leads B for counterclockwise rotation when viewed from the shaft side of the encoder.

### Electrical

- Specifications apply over the entire operating temperature range.
- Typical values are specified at  $V_{CC} = 5.0V_{DC}$  and  $25^{\circ}C$ .
- For complete details, see the [EM1<sup>\(5\)</sup>](#) and [EM2<sup>\(6\)</sup>](#) product pages.

| Parameter                  | Min.       | Typ.           | Max.           | Units               | Conditions                                                                                                         |
|----------------------------|------------|----------------|----------------|---------------------|--------------------------------------------------------------------------------------------------------------------|
| Supply Voltage             | 4.5        | 5.0            | 5.5            | V                   |                                                                                                                    |
| Supply Current             |            | 27<br>54<br>72 | 33<br>62<br>85 | mA<br>mA<br>mA      | CPR < 1000, no load<br>CPR ≥ 1000 and < 3600, no load<br>CPR ≥ 3600, no load                                       |
| Low-level Output           |            | 0.05<br>0.25   | 0.5<br>0.5     | V<br>mA<br>mA<br>mA | $I_{OL} = 8mA$ max., CPR < 3600<br>$I_{OL} = 5mA$ max., CPR ≥ 3600<br>no load, CPR < 3600<br>no load, CPR ≥ 3600   |
| High-level Output          | 2.0<br>2.0 | 4.8<br>3.5     |                | V<br>V<br>V<br>V    | $I_{OH} = -8mA$ max., CPR < 3600<br>$I_{OH} = -5mA$ max., CPR ≥ 3600<br>no load, CPR < 3600<br>no load, CPR ≥ 3600 |
| Output Current Per Channel | -8<br>-5   |                | 8<br>5         | mA<br>mA            | CPR < 3600<br>CPR ≥ 3600                                                                                           |
| Output Rise Time           |            | 110<br>50      |                | nS<br>nS            | CPR < 3600<br>CPR ≥ 3600                                                                                           |
| Output Fall Time           |            | 35<br>50       |                | nS<br>nS            | CPR < 3600<br>CPR ≥ 3600                                                                                           |

# Ball Bearing Rotary Shaft Encoder

Product ID | H3



## Specifications

### Pin-Outs

| Pin | Description |
|-----|-------------|
| 1   | Ground      |
| 2   | Index       |
| 3   | A channel   |
| 4   | +5VDC power |
| 5   | B channel   |

## Notes

- Cables and connectors are not included and must be ordered separately.
- US Digital® warrants its products against defects in materials and workmanship for two years. See complete [warranty<sup>\(7\)</sup>](#) for details.

# Ball Bearing Rotary Shaft Encoder

Product ID | H3



## Part Number Guide and Configuration Options

|                              |                |             |         |
|------------------------------|----------------|-------------|---------|
| Product ID - [ ] - [ ] - [ ] |                |             |         |
| H3                           | CPR            | Index       | Housing |
| 64                           | IE (Index)     | D (Default) |         |
| 100                          | NE (Non-Index) |             |         |
| 200                          |                |             |         |
| 400                          |                |             |         |
| 500                          |                |             |         |
| 512                          |                |             |         |
| 1000                         |                |             |         |
| 1024                         |                |             |         |
| 1800                         |                |             |         |
| 2000                         |                |             |         |
| 2048                         |                |             |         |
| 2500                         |                |             |         |
| 3600                         |                |             |         |
| 4000                         |                |             |         |
| 4096                         |                |             |         |
| 5000                         |                |             |         |
| 7200                         |                |             |         |
| 8000                         |                |             |         |
| 8192                         |                |             |         |
| 10000                        |                |             |         |

**PLEASE NOTE:** This chart is for informational use only. Certain product configuration combinations are not available. Visit the [H3 product page<sup>\(8\)</sup>](#) for pricing and additional information.

# Ball Bearing Rotary Shaft Encoder

Product ID | H3



## Reference Links



(1) H6 Ball Bearing Rotary Shaft Encoder



(2) PC4 Incremental Encoder Cable Driver



(3) PC5 Broadcom/Avago Compatible Incremental Encoder Cable Driver



(4) Shaft and Bore Tolerances - TB1001 Technical Bulletin



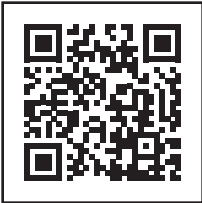
(5) EM1 Transmissive Optical Encoder Module



(6) EM2 Transmissive Optical Encoder Module



(7) Legal Terms and Conditions, Warranty



(8) H3 Ball Bearing Rotary Shaft Encoder